PRACTICE SHEET-1

1. A train started from a station with a certain number of passengers. At the first halt, 1/3rd of its passengers got down and 120 passengers got in. At the second halt, half of the passengers got down and 100 persons got in. Then, the train left for its destination with 240 passengers. How many passengers were there in the train when it started?
(a) 540 (b) 480
(c) 360 (d) 240

2. A person bought a certain number of books for Rs 80. If he had bought 4 more books for the same sum, each book would have cost Rs 1 less. What is the price of each book?
(a) Rs 10 (b) Rs 8
(c) Rs 5 (d) Rs 4

3. A person bought 5 tickets from a station *P* to a station *Q* and 10 tickets from the station *P* to a station *R*. He paid Rs 350. If the sum of a ticket from *P* to *Q* and a ticket from *P* to *R* is Rs 42, then what is the fare from *P* to *Q*?

7. Let there be three simultaneous linear equations in two unknowns, which are non-parallel and non-collinear them. What can be the number of solutions (if they do exist)?
(a) One or infinite (b) Only one
(c) Exactly two (d) exactly three

8. If 1 is added to the denominator of a fraction, it becomes 1/2 and if 1 is added to the numerator, the fraction becomes 1. What is the fraction?
(a) 5/9 (b) 2/3
(c) 4/7 (d) 10/11

9. If $\frac{2}{x}+\frac{3}{y}=\frac{9}{xy}$ and $\frac{4}{x}+\frac{9}{y}=\frac{21}{xy}$, where $x \neq 0$ and $y \neq 0$, then what is the value of $x + y$?
(a) 2 (b) 3
(c) 4 (d) 8

10. The sum of two numbers is 80. If the larger